Manzil JEE - Sheet (2025)

Quadratic Equations

Most Important JEE PYQs

Single Correct Type Questions

1. Let α, β be the roots of the quadratic equation $x^2+\sqrt{6}x+3=0$. Then $\dfrac{\alpha^{23}+\beta^{23}+\alpha^{14}+\beta^{14}}{\alpha^{15}+\beta^{15}+\alpha^{10}+\beta^{10}}$ is equal to

[12 April, 2023 (Shift-I)]

(*a*) 729 (*b*) 72
(*c*) 81 (*d*) 9

2. Let α, β be the roots of the equation $x^2-\sqrt{2}x+2=0$. Then $\alpha^{14}+\beta^{14}$ is equal to **[13 April, 2023 (Shift-II)]**

(*a*) $-64\sqrt{2}$ (*b*) $-128\sqrt{2}$
(*c*) -64 (*d*) -128

3. The sum of all the roots of the equation $|x^2-8x+15|-2x+7=0$ is: **[6 April, 2023 (Shift-I)]**

(*a*) $9+\sqrt{3}$ (*b*) $11+\sqrt{3}$
(*c*) $9-\sqrt{3}$ (*d*) $11-\sqrt{3}$

4. Let $f(x)$ be a quadratic polynomial such that $f(-2)+f(3)=0$. If one of the roots of $f(x)=0$ is -1, then the sum of the roots of $f(x)=0$ is equal to: **[28 June, 2022 (Shift-II)]**

(*a*) $\dfrac{11}{3}$ (*b*) $\dfrac{7}{3}$
(*c*) $\dfrac{13}{3}$ (*d*) $\dfrac{14}{3}$

5. If α, β are the roots of the equation

$$x^2-\left(5+3^{\sqrt{\log_3 5}}-5^{\sqrt{\log_5 3}}\right)x+3\left(3^{(\log_3 5)^{\frac{1}{3}}}-5^{(\log_5 3)^{\frac{2}{3}}}-1\right)=0$$

then the equation, whose roots are $\alpha+\dfrac{1}{\beta}$ and $\beta+\dfrac{1}{\alpha}$,

[27 July, 2022 (Shift-II)]

(*a*) $3x^2-20x-12=0$
(*b*) $3x^2-10x-4=0$
(*c*) $3x^2-10x+2=0$
(*d*) $3x^2-20x+16=0$

6. Let α and β be the roots of the equation $x^2+(2i-1)=0$. Then, the value of $|\alpha^8+\beta^8|$ is equal to :

[29 June, 2022 (Shift-I)]

(*a*) 50 (*b*) 250
(*c*) 1250 (*d*) 1500

7. Let α and β be the roots of $x^2-6x-2=0$. If $a_n=\alpha^n-\beta^n$ for $n\geq 1$, then the value of $\dfrac{a_{10}-2a_8}{3a_9}$ is:

[25 Feb, 2021 (Shift-II)]

(*a*) 2 (*b*) 1 (*c*) 4 (*d*) 3

8. The number of real roots of the equation $e^{6x}-e^{4x}-2e^{3x}-12e^{2x}+e^x+1=0$ is:

[25 July, 2021 (Shift-I)]

(*a*) 1 (*b*) 4 (*c*) 2 (*d*) 6

9. If α and β are the distinct roots of the equation $x^2+(3)^{1/4}x+3^{1/2}=0$, then the value of $\alpha^{96}(\alpha^{12}-1)+\beta^{96}(\beta^{12}-1)$ is equal to: **[20 July, 2021 (Shift-I)]**

(*a*) 56×3^{25} (*b*) 28×3^{25} (*c*) 52×3^{24} (*d*) 56×3^{24}

10. The value of $4+\cfrac{1}{5+\cfrac{1}{4+\cfrac{1}{5+\cfrac{1}{4+\dots\infty}}}}$ **[17 March, 2021 (Shift-I)]**

(*a*) $2+\dfrac{2}{5}\sqrt{30}$ (*b*) $2+\dfrac{4}{\sqrt{5}}\sqrt{30}$
(*c*) $4+\dfrac{4}{\sqrt{5}}\sqrt{30}$ (*d*) $5+\dfrac{2}{5}\sqrt{30}$

11. cosec 18° is a root of the equation:

[31 Aug, 2021 (Shift-I)]

(*a*) $x^2+2x-4=0$
(*b*) $x^2-2x+4=0$
(*c*) $4x^2+2x-1=0$
(*d*) $x^2-2x-4=0$

12. Let a, b, c be in arithmetic progression. Let the centroid of the triangle with vertices (a, c), $(2, b)$ and (a, b) be $\left(\frac{10}{3}, \frac{7}{3}\right)$. If α, β are the roots of the equation $ax^2 + bx + 1 = 0$, then the value of $\alpha^2 + \beta^2 - \alpha\beta$ is: **[24 Feb, 2021(Shift-II)]**

(*a*) $-\frac{69}{256}$ (*b*) $\frac{69}{256}$ (*c*) $\frac{27}{256}$ (*d*) $-\frac{71}{256}$

13. If α and β be two roots of the equation $x^2 - 64x + 256 = 0$. Then the value of $\left(\frac{\alpha^3}{\beta^5}\right)^{1/8} + \left(\frac{\beta^3}{\alpha^5}\right)^{1/8}$ is: **[6 Sep, 2020 (Shift-I)]**

(*a*) 3 (*b*) 2 (*c*) 4 (*d*) 1

14. Let α and β be the roots of the equation, $5x^2 + 6x - 2 = 0$. If $S_n = \alpha^n + \beta^n$, $n = 1, 2, 3.....$, then **[2 Sep, 2020 (Shift I)]**

(*a*) $6S_6 + 5S_5 = 2S_4$ (*b*) $6S_6 + 5S_5 + 2S_4 = 0$
(*c*) $5S_6 + 6S_5 = 2S_4$ (*d*) $6S_6 + 6S_5 + 2S_4 = 0$

15. Let α and β be the roots of the equation $x^2 - x - 1 = 0$. If $p_k = (\alpha)^k + (\beta)^k$, $k \geq 1$, then which of the following statements is not true? **[7 Jan, 2020 (Shift-II)]**

(*a*) $p_5 = p_2 \cdot p_3$
(*b*) $(p_1 + p_2 + p_3 + p_4 + p_5) = 26$
(*c*) $p_3 = p_5 - p_4$
(*d*) $p_5 = 11$

16. Let S be the set of all real roots of the equation, $3^x(3^x - 1) + 2 = |3^x - 1| + |3^x - 2|$. Then S: **[8 Jan, 2020 (Shift-II)]**

(*a*) contains exactly two elements
(*b*) is a singleton
(*c*) contains at least four elements
(*d*) is an empty set

17. Let α and β be the roots of the quadratic equation $x^2 \sin\theta - x(\sin\theta\cos\theta + 1) + \cos\theta = 0$ $(0 < \theta < 45^0)$, and $\alpha < \beta$. Then $\sum_{n=0}^{\infty}\left(\alpha^n + \frac{(-1)^n}{\beta^n}\right)$ is equal to: **[11 Jan, 2019 (Shift-II)]**

(*a*) $\frac{1}{1-\cos\theta} - \frac{1}{1+\sin\theta}$ (*b*) $\frac{1}{1+\cos\theta} + \frac{1}{1-\sin\theta}$
(*c*) $\frac{1}{1-\cos\theta} + \frac{1}{1+\sin\theta}$ (*d*) $\frac{1}{1+\cos\theta} - \frac{1}{1-\sin\theta}$

18. Let α and β be two roots of the equation $x^2 + 2x + 2 = 0$, then $\alpha^{15} + \beta^{15}$ is equal to: **[9 Jan, 2019 (Shift-I)]**

(*a*) -256 (*b*) 512
(*c*) -512 (*d*) 256

19. Let α and β be the roots of $x^2 - x - 1 = 0$, with $\alpha > \beta$. For all positive integers n, define $a^n = \frac{\alpha^n - \beta^n}{\alpha - \beta}$, $n \geq 1$ **[JEE Advanced-2019]**

$b_1 = 1$ and $b_n = a_{n-1} + a_{n+1}$, $n \geq 2$.
Then which of the following options is/are correct?

(*a*) $a_1 + a_2 + a_3 + ... + a_n = a_{n+2} - 1$ for all $n \geq 1$
(*b*) $\sum_{n=1}^{\infty}\frac{a_n}{10^n} = \frac{10}{89}$
(*c*) $\sum_{n=1}^{\infty}\frac{b_n}{10^n} = \frac{8}{89}$
(*d*) $b_n = \gamma^n + \beta^n$ for all $n \geq 1$

20. Consider the quadratic equation $(c-5)x^2 - 2cx + (c-4) = 0$, $c \neq 5$. Let S be the set of all integral values of c for which one root of the equation lies in the interval (0, 2) and its other root lies in the interval (2, 3). Then the number of elements in S is: **[10 Jan, 2019 (Shift-I)]**

(*a*) 18 (*b*) 12
(*c*) 10 (*d*) 11

21. Let

$$S = \left\{x : x \in \mathsf{R} \text{ and } (\sqrt{3}+\sqrt{2})^{x^2-4} + (\sqrt{3}-\sqrt{2})^{x^2-4} = 10\right\}$$

Then $n(S)$ is equal to **[1 Feb, 2023 (Shift-I)]**

(*a*) 2
(*b*) 4
(*c*) 6
(*d*) 0

22. The equation $x^2 - 4x + [x] + 3 = x[x]$, where $[x]$ denotes the greatest integer function, has: **[24 Jan, 2023 (Shift-I)]**

(*a*) exactly two solutions in $(-\infty, \infty)$
(*b*) no solution
(*c*) a unique solution in $(-\infty, 1)$
(*d*) a unique solution in $(-\infty, \infty)$

Integer Type Questions

23. Let $a \in R$ and let α, β be the roots of the equation $x^2 + 60^{\frac{1}{4}}x + a = 0$. If $\alpha^4 + \beta^4 = -30$, then the product of all possible values of a is ___________.

[25 Jan, 2023 (Shift-II)]

24. Let m and n be the numbers of real roots of the quadratic equations $x^2 - 12x + [x] + 31 = 0$ and $x^2 - 5|x + 2| - 4 = 0$ respectively, where $[x]$ denotes the greatest integer $\leq x$. Then $m^2 + mn + n^2$ is equal to ____.

[8 April, 2023 (Shift-II)]

25. Let p and q be two real numbers such that $p + q = 3$ and $p^4 + q^4 = 369$. Then $\left(\frac{1}{p}+\frac{1}{q}\right)^{-2}$ is equal to _________.

[26 June, 2022 (Shift-II)]

26. The number of real solutions of the equation $e^{4x} + 4e^{3x} - 58e^{2x} + 4e^{x} + 1 = 0$ is **[28 June, 2022 (Shift-I)]**

27. The number of distinct real roots of the equation $3x^4 + 4x^3 - 12x^2 + 4 = 0$ is _____. **[27 Aug, 2021 (Shift-I)]**

28. Let α and β be two real numbers such that $\alpha + \beta = 1$ and $\alpha\beta = -1$. Let $P_n = (\alpha)^n + (\beta)^n$, $P_{n-1} = 11$ and $P_{n+1} = 29$ for some integer $n \geq 1$. Then, the value of P_n^2 is_____

[26 Feb, 2021 (Shift-II)]

29. Let $f(x)$ be a quadratic polynomial with leading coefficient 1 such that $f(0) = p$, $p \neq 0$, and $f(1) = \frac{1}{3}$. If the equations $f(x) = 0$ and fo fo fo $f(x) = 0$ have a common real root, then $f(-3)$ is equal to _______ **[25 July, 2022 (Shift-II)]**

30. The number of distinct real roots of the equation $x^5(x^3 - x^2 - x + 1) + x(3x^3 - 4x^2 - 2x + 4) - 1 = 0$ is.........

[26 July, 2022 (Shift-I)]